

MOORED BUOY 21002, Japan

WMO: 992260

Lat: 37.700N

Lon: 134.300E

Elev: 0

StdP: 101.33

Time Zone: 9.00 (E09)

Period: 90-00

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-2.2	-1.0	-6.4	2.2	2.9	-5.2	2.4	3.0	22.3	2.0	20.7	3.8	12.4	0	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	1.6	28.1	26.0	27.4	25.5	26.8	24.9	26.5	27.5	25.9	27.2	25.3	26.7	6.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	21.5	27.3	25.5	20.7	27.0	24.8	19.9	26.5	82.7	27.6	80.4	27.4	77.9	26.7	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.6	16.7	14.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.8	4.4	3.9	6.5	10.2	14.2	18.1	22.3	25.4	22.5	17.8	12.6	7.7		
	DBStd	7.57	3.04	3.14	2.67	2.44	2.07	1.69	2.07	1.46	2.01	2.53	3.38	3.29		
	HDD10.0	589	176	172	112	25	1	0	0	0	0	0	17	87		
	HDD18.3	2145	433	405	368	243	128	23	1	0	1	39	173	331		
	CDD10.0	1994	1	1	3	32	132	243	381	478	374	242	93	15		
	CDD18.3	507	0	0	0	0	1	16	123	220	125	22	0	0		
	CDH23.3	2353	0	0	0	0	0	0	331	1651	366	5	0	0		
	CDH26.7	154	0	0	0	0	0	0	3	138	12	0	0	0		
(14) Wind	WSAvg	7.1	9.1	8.7	7.2	6.4	5.6	5.2	5.8	5.9	6.8	7.2	8.0	9.0		
(15) Precipitation	PrecAvg	1022	49	42	48	76	88	93	131	133	150	92	68	51		
	PrecMax	1369	104	81	96	140	182	226	386	395	261	167	144	112		
	PrecMin	711	20	9	4	9	28	24	15	39	35	19	26	12		
	PrecStd	144	19	18	22	37	36	50	87	76	61	35	27	21		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	13.7	12.6	15.9	19.5	23.0	27.2	29.2	27.7	23.9	19.4	16.2		
		MCWB	9.8	12.3	10.7	14.6	17.2	21.1	24.8	26.1	25.2	22.4	16.0	13.7		
	2%	DB	11.4	11.0	11.9	14.9	18.5	22.2	26.4	28.5	27.0	22.4	18.6	15.3		
		MCWB	8.6	9.1	10.1	13.4	16.7	20.7	24.4	26.1	24.8	19.6	15.3	12.5		
	5%	DB	10.1	9.8	11.2	14.1	17.5	21.2	25.8	28.0	26.1	21.6	17.9	14.0		
		MCWB	7.3	7.7	9.1	12.7	16.1	19.8	24.1	26.1	23.3	18.3	14.7	10.5		
	10%	DB	8.9	8.7	10.5	13.2	16.9	20.5	25.1	27.4	25.2	21.1	17.2	12.7		
		MCWB	6.1	6.2	8.4	11.7	15.6	19.1	23.5	25.6	22.7	17.6	13.9	9.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.3	13.4	11.2	15.3	18.1	21.6	25.7	27.2	26.4	23.0	17.1	15.2		
		MCDB	12.8	14.4	12.1	15.8	18.7	22.5	26.6	27.8	27.1	24.3	18.3	16.0		
	2%	WB	9.1	10.0	10.6	13.7	17.2	21.0	25.0	26.8	25.4	21.1	16.4	12.3		
		MCDB	11.3	10.7	11.7	14.6	17.9	22.2	26.1	27.7	26.6	22.0	18.0	14.9		
	5%	WB	7.7	8.0	9.7	13.0	16.6	20.1	24.5	26.4	24.3	19.9	15.4	10.6		
		MCDB	10.0	9.8	11.0	14.0	17.3	21.0	25.6	27.5	25.6	21.1	17.3	13.2		
	10%	WB	6.6	6.3	8.9	12.2	15.9	19.5	23.9	26.0	23.6	18.5	14.3	9.4		
		MCDB	8.8	8.6	10.4	13.2	16.7	20.4	25.0	27.3	25.0	20.8	16.8	12.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.8	3.5	3.3	2.4	2.1	1.6	1.5	1.6	1.8	2.7	3.7	4.0		
		MCDBR	5.7	4.8	4.0	2.3	2.1	1.6	1.8	1.8	2.2	3.0	4.7	4.7		
	5% WB	MCWBR	5.7	4.9	4.6	2.3	2.0	1.7	1.4	1.7	2.7	4.0	5.2	5.0		
		MCWBR	5.9	4.2	4.0	2.1	2.3	1.6	1.5	1.5	2.0	3.6	4.4	5.2		
(40) Clear-Sky Solar Irradiance	taub		0.431	0.462	0.526	0.553	0.556	0.555	0.562	0.509	0.457	0.446	0.437	0.437		
	taud		2.012	1.946	1.825	1.812	1.875	1.962	2.011	2.178	2.265	2.141	2.081	1.996		
	Ebn at Noon		705	740	732	743	750	750	741	773	794	752	696	661		
	Edh at Noon		130	157	195	209	200	183	173	144	125	128	121	124		
(44) All-Sky Solar Radiation	RadAvg		1.55	2.50	3.91	5.07	5.47	5.38	5.01	5.28	4.22	3.32	2.02	1.37		
	RadStd		0.17	0.23	0.29	0.43	0.52	0.41	0.57	0.57	0.34	0.27	0.20	0.13		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page